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PROJECT REPORT

ON

"AUTOMATED WEAPON DETECTION AND EVIDENCE FILTERING SYSTEM"

Submitted To

Department of Cyber Security & Digital Forensics

National Forensic Sciences University

Submitted By

ANSHUL DHANJWAL

(022200300004037)

Under the Supervision of

Dr. BRAHM PRAKASH

Department Of Cyber Security & Digital Forensics

National Forensic Sciences University,

Delhi Campus, New Delhi – 110085, India

DECLARATION

I hereby declare that the thesis entitled "AUTOMATED WEAPON DETECTION AND EVIDENCE FILTERING SYSTEM" is a research work done by me and no part of the thesis has been presented earlier and will be presented for any degree, diploma or similar title at any other institute/university.

ANSHUL DHANJWAL
022200300004037
BTECH-MTECH(CSE)
2022-27

Date: 18/12/2025
Place: NFSU, DELHI

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This is to certify that the work contained in the dissertation entitled **"AUTOMATED WEAPON DETECTION AND EVIDENCE FILTERING SYSTEM"**, submitted by **ANSHUL DHANJWAL (Enroll No.: 022200300004037)** for the award of the degree of **B.Tech - M.Tech Computer Science & Engineering (Cyber Security)**to the **National Forensic Sciences University**, **Delhi Campus**, is a record of bonafide research works carried out by him/her under my supervision and guidance.

Dr. BRAHM PRAKASH
Assistant Professor
School of Cyber Security & Digital Forensics
National Forensic Sciences University
Delhi Campus, Delhi, India

Date:18/12/2025
Place: NFSU, DELHI

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With Sincere Regards,

ANSHUL DHANJWAL

022200300004037

BTECH-MTECH(CSE)

2022-27

ABSTRACT

The rapid advancement of crime scene investigation methodologies necessitates the development of intelligent, automated systems to assist forensic experts in analyzing large volumes of evidence effectively. This project presents an **Automated Weapon Detection and Evidence Filtering System (AWDEFS)**, a deep learning-based solution leveraging YOLOv8 (You Only Look Once version 8) architecture for real-time detection and classification of weapons in digital forensic imagery.

The system is designed to address the critical challenge faced by forensic investigators: the time-consuming manual process of sifting through hundreds of crime scene photographs to identify those containing weapons of interest. AWDEFS automates this triage process, allowing investigators to filter bulk image collections by specific weapon classes (firearms, knives, and other weapons) within minutes, significantly reducing investigation time while maintaining forensic integrity.

The project implements a complete pipeline comprising data acquisition, model training using YOLOv8-nano architecture optimized for CPU inference, and a flexible filtering system with class-based selection. The system was trained on 2000+ images with 30e00 annotated weapon instances, utilizing a 30-epoch optimization scheme that completes in approximately 8-10 days on standard CPU hardware, making it accessible for deployment in resource-constrained forensic laboratories.

Key contributions include: (1) A production-ready Python implementation utilizing the Ultralytics framework; (2) Flexible command-line interface supporting selective filtering by weapon class; (3) Automated report generation (CSV and summary statistics) for forensic documentation; (4) Annotated output with visual bounding boxes and confidence scores for expert verification; (5) Modular architecture enabling future extensions to bloodstain detection, personal identification, and multi-modal evidence analysis.

Experimental results demonstrate 5 firearm detections and 4 knife detections across 7 filtered images from a test set of 11 images, validating the system's capability to accurately identify and extract relevant evidence. The system achieves a balance between detection accuracy (precision) and speed, making it a practical tool for real-world forensic applications. Future enhancements will focus on multi-source dataset integration, real-time video stream processing, and integration with existing forensic laboratory management systems (LIMS).

Keywords: Deep Learning, Object Detection, YOLOv8, Weapon Detection, Forensic Science, Evidence Filtering, Computer Vision, Automated Triage, Criminal Investigation, Crime Scene Analysis.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
CSI	Crime Scene Investigation
DL	Deep Learning
DNN	Deep Neural Network
FN	False Negative
FP	False Positive
GPU	Graphics Processing Unit
HOG	Histogram of Oriented Gradients
IoU	Intersection over Union
LIMS	Laboratory Information Management System
mAP	mean Average Precision
ML	Machine Learning
NMS	Non-Maximum Suppression
NLP	Natural Language Processing
NFSU	National Forensic Sciences University
NNS	Nearest Neighbor Search
ONNX	Open Neural Network Exchange
ReLU	Rectified Linear Unit
RGB	Red, Green, Blue
SGD	Stochastic Gradient Descent
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
YOLO	You Only Look Once
AWDEFS	Automated Weapon Detection and Evidence Filtering System
DCNN	Deep Convolutional Neural Network
DPI	Dots Per Inch

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CHAPTER 1: INTRODUCTION

1.1 Introduction and Problem Summary

Crime scene investigation has changed a lot in the last twenty years or so, mostly because of new forensic tech and digital stuff coming into play. But even with that, going through all the photos from a scene by hand is still one of the biggest time sinks. Investigators end up with hundreds, sometimes thousands of pictures from just one case, and they have to look at each one to spot things like weapons or bloodstains or whatever evidence matters. That part alone eats up so much time and effort.

I think the real problem shows up in how digital evidence is handled these days. From what I read in some research papers and other standards, this manual reviewing has turned into a major bottleneck for a lot of investigations. Like, a regular crime scene might give you 500 to 1000 photos that need at least a basic check, plus all sorts of evidence to sort out, from weapons to biological traces or tiny bits of stuff. And the images are not always great, with different lighting and quality issues. Then there's the whole chain of custody thing, making sure everything is documented properly. All that manual work can take anywhere from 8 to 12 hours for one investigation.

It feels like manual analysis just invites problems too. For one, human error creeps in because people get tired and miss important details. Or you waste hours on photos that don't even matter. Different investigators might categorize things inconsistently, which is kind of a bias issue. It pulls experts away from the bigger picture analysis they should be doing. And rushing through to save time could mess with how thorough the evidence check is, raising concerns about integrity. Some people might argue its not that bad, but it seems inefficient overall.

Role of Technology in Forensic Science

Technology is changing forensic science a lot these days. Artificial intelligence and computer vision are getting mixed in, and it feels like a big change toward using data more in investigations. I mean, machine learning stuff, especially the deep learning kinds, they handle recognizing patterns in images pretty well.

Take face recognition for identifying people in facial forensics. Or detecting if documents are forged, like in questioned documents cases. Fingerprint matching is another one, analyzing those prints. Then there's identifying biological stains. And spotting behavioural patterns in surveillance videos.

It seems like this deep learning success shows that smart automation can really help out in forensics. It doesn't replace the experts, but it adds to what they do and speeds up decisions. Kind of augments the whole process. Some might think it could take over, but I don't see it that way yet. The evidence points to it enhancing things without fully changing everything.

Weapon Detection in Forensic Context

Identifying weapons is critical at a crime scene for several reasons:

1. Evidence – Evaluation of the weapon and its surrounding environment can provide information concerning the events that took place prior to the violent crime.
2. Safety – Weapons must be handled safely in order to ensure a safe investigation. By identifying weapons quickly, investigators can safely handle evidence.
3. Investigative Direction – Identifying a weapon type will assist in developing a suspect profile and directing the investigative focus.
4. Crime Statistics – Weapons are utilized in approximately 45% of violent crimes. Therefore, accurate identification of a weapon & type will aid in the investigation process.
5. Ballistics – For example, after firearms are identified, the forensic ballistic expert may access ballistic data on firearms for future investigations.

Project Motivation and Significance

With the completion of this project, two specific purposes were achieved:

1. The Desirable Forensic Technology: A practical, implementable system has been created to meet an actual need in forensic labs.
2. The Demonstration of AI Use in Police Work: A proof of concept for intelligent evidence filtering in limited resource settings was prepared.
3. The Potential for Use by a Widespread Audience: An automated system made to work for all CPUs in a forensic laboratory because of a lack of built-in graphics processing unit's capabilities.
4. The Feasibility of Operating the System in Forensic Lab Workflows: Created with a training duration of 30 to 40 minutes which can be reasonably incorporated into planned workflow in forensic laboratories.

The Automated Weapon Detection and Evidence Filtering System (AWDEFS) is intended to provide a way of bridging the gap between what is possible with cutting edge computers and what is practically necessary in forensic laboratories.

1.2 Aim and Objectives of the Project

Primary Aim

The focus of the project is to build a system using deep learning technology to help law enforcement with quickly and efficiently identifying and collecting critical evidence from crime scene photographs. The project will also support law enforcement agencies in maintaining the integrity of investigations and meeting the requirements of legal and forensic protocols.

Specific Objectives

Goal 1: Design and Build a Functional Weapon Detection System

● Create and develop a working weapon detection system based on YOLOv8 (You Only Look Once version 8) Deep Learning Model

● Provide system performance that satisfies use in real-time or near real-time forensic laboratory operations

● Provide CPU-compatible operation to maximize use in low-resource environments

Goal 2: Provide Flexible and Usable System

• Create a flexible, class-based weapon filtering system to allow investigators to extract images based on weapon classes.

● Create a user-friendly command line interface that allows investigators to granularly filter images based on weapon class.

● Allow Investigators to specify their desired outputs (i.e., filtered image set, annotated images, summary reports, etc.)

Goal 3: Forensic Compliance

• Create forensic-compliant documentation as follows:

Goal 4: System Performance Testing

● Determine a baseline of weapon detection accuracy for the weapon detection system.

● Evaluate how the system performs with various image qualities and lighting conditions.

● Record precision, recall, and F1-score metrics for each weapon class.

Identify limiting factors and potential failure modes for the weapon detection system, and identify areas to address in future improvements.

Goal 5: Scalability and Future-Proofing

● Develop and implement a modular architecture to allow future weapon detection system extensions.

Goal 6: Documentation and Knowledge Transfer

● Produce comprehensive technical documentation.

● Provide Guidelines for Forensic Practitioners.

● Document system limitations and potential applications of the system.

● Create a base for future development of weapon detection research and systems.

1.3 Scope of the Project

What is Included

System Scope:

1. Object Detection Component: YOLOv8-based neural network to identify the weapon in an image.
2. Image Filtering Pipeline: To automatically pull all images of the specified weapon category(s) by the user.
3. Reporting System: To create csv files and summary statistics for use in investigation files.
4. Annotated Image System: To create annotated images with bounding boxes and confidence levels.
5. User Interface: A command-line interface that allows for user-defined filtering parameters.

Data Scope:

1. Training Data: The weapon-Kaggle dataset contains 2000+ images and 3000+ classified instances.
2. Supported File Formats: JPG and PNG image files (standard forensic photo file types)
3. Image Resolutions: Images can be filtered from 416 x 416-resolutions to full-quality filtered images.

Functional Scope:

1. Allows for batch processing of entire folders of images.
2. User-defined filtering by weapon type (one or more types).
3. Ability to create annotated images as requested by the user.
4. Automatically determines performance metrics.
5. Creates summary report files.

What is NOT Included

Excluded From Project's Scope:

1.) The system is only designed for static image analysis. Therefore, real-time video processing and development of an appropriate module is considered in future phases of the project.
2.) The ability to identify individuals based on their facial recognition in photographs taken at crime scenes has been explicitly excluded from this project due to ethical and privacy concerns.
3.) 3D reconstruction of crime scenes has also been excluded from the scope of this project.
4.) The ability to connect to or use ballistics databases, weapons registries, or AFIS systems will be addressed at future phases of the project.
5.) This project will only focus on the detection of weapons; other types of evidence, such as bloodstains, fingerprints, and trace evidence are not included within this project's scope.
6.) Although it will assist investigators by providing relevant evidence, this system does not have the ability to make decisions autonomously; therefore, it will be up to the investigator to assess final evidence for admissibility, etc.
7.) While formats have been established to support documentation related to the preservation of evidence and the creation of a chain-of-custody, the integration of this system with an overall LIMS database will be addressed at future phases of the project.

Limitations and Constraints

Technical Limitations:
1. The limits of training data: Limited amount of training data (2000+ images) means less generalization across the entire range of possible weapon types and conditions.
2. Types of weapons: The system only trained to recognize handguns and knives; it does not recognize other categories of weapons.
3. The effects of lighting and occlusion on performance: Performance of the system degrades under extremely bright or dark lighting or when significant parts of a weapon are blocked from view.
4. Quality of input images: The performance of the system is dependent on the quality of the input images, such as resolution, focus, and lighting.

Operational Constraints:
1. Time to train a complete model: Trains a complete model in approximately 30-40 minutes on CPU-based hardware.

2. Speed of processing: The time to perform inference for a single image ranges from approximately 100-200 ms (depending on the image resolution and CPU hardware).
3. Required amount of memory: The system will typically require approximately 4 GB of RAM to operate at maximum efficiency.
4. The model is trained only to look for specific weapon appearances and may require retraining to recognize the appearance of novel weapon designs.

Scope Constraints:
1. The inability to perform real-time image processing: The application was designed to process a batch collections of images, and is not designed to monitor activity in real-time.
2. The inability to support the autonomous decision-making process: The system is capable of automatically filtering results based on its detection capability; however the investigators will still be required to verify the results.
3. Limitations of single-image processing: All images will be evaluated individually; there is no temporal relationship between sequences of images.
4. The lack of integration capability with external systems: The application is standalone, and developing an integrating solution between this application and LIMS or other systems will require additional development efforts.

Deliverables

The project delivers:

1. Software System
2. Documentation
3. Experimental Results
4. Future Development Foundation

CHAPTER 2: LITERATURE SURVEY

2.1 Current/Existing System

2.1.1 Study of Current System

Manual Evidence Analysis Process

Forensic Evidence Analysis within a Forensic Lab includes the assessment of the evidence manually and utilizing human judgement to categorize the evidence.

The typical steps for assessing evidence include:

1. Collection of Evidence
2. Sorting Evidence manually (in an initial sort)
3. Documenting Evidence
4. Continuing to investigate based upon the material that has been sorted and documented.

Manual Assessment Results -

1. Human Fatigue - While many people may be capable of reviewing hundreds of images, fatigue impacts your ability to focus and concentrate at that level.
2. Inconsistent Application of Criteria - Different people (technicians) will have different criteria for categorizing evidence and this inconsistency makes it difficult to assess the condition of the evidence and develop an appropriate investigative strategy.
3. Misuse of Resources - Hiring highly specialized and trained professionals to conduct basic categorization of evidence wastes resources, both in terms of money and time.
4. Time - The majority of the investigation consists of "mechanical" filtering.
5. Missed Evidence - There is a large amount of evidence that is collected, therefore, subtle evidence is often overlooked due to the volume of the evidence.

Current Technology Adoption

Some progressive forensic laboratories have adopted technology-assisted approaches:

1. Image Management Software (LIMS systems)

2. Computer Vision Research Systems

3. Surveillance Video Analysis Tools

4. Limited Automation

2.1.2 Problems & Weakness of Current System

Identified Deficiencies

1. Inefficient Use of Time

- The average time spent triaging per-scene evidence ranges from 1 to 3 hours each time this occurs. This can add up very quickly as there are over 1,000 investigations performed every year resulting in significant resources used.
- Opportunity Cost: If experts aren't spending their valuable time at higher levels, they have lost the opportunity to make that contribution elsewhere.
- Eventually, artificial timelines will be created if the above is not considered when making conclusions.

2. Variations in Quality

- Each technician produces a different "number of weapons per-image" resulting from the standards they use (not because they don't follow the same procedure).
- There is an increased chance of a "borderline" piece of evidence being missed.

3. Limitations Due to Cognitive Fatigue

- Humans' ability to concentrate significantly decreases the longer they are engaged in a particular task or activity.
- This reduces the ability for a person to identify subtle evidence due to image fatigue.
- The lack of objective standards to evaluate whether evidence has been adequately reviewed.

4. Scalability Issues with Problem

- Special Events like Mass Casualty Incidents with multiple crime scenes creates the potential for overwhelming the Triage System capacity.

5. Forensic Integrity Problems with Problem

- Manual methods to collect evidence are vulnerable to bias based on an investigators - preconceived notions
- Incomplete documentation of exclusion criteria or lack of documentation at all

problems exist when reproducing decisions made regarding patient assignments
- The madness surrounding chain of custody increases when humans are involved.

6. Cost and Resources Impacted with Problem
Costs associated with;
- Personnel Costs (monthly). (Manual Evaluation) = (Tech Hours × Cost Per Hour).
- Opportunity Cost (delayed investigations).
- Training New Employees on Manual Categorisation.
- Variable Quality (Requires Supervisor Review).

2.2 Requirements of New System

Functional Requirements

FR1: Weapon Discovery

●Tool will identify and locate all types of weapons (knives, pistols, revolvers, etc.) from digital photo collections

●Tool will supply confidence levels associated with each identification

●Tool will provide X-Y coordinates for plotting the detection in space through bounding boxes

FR2: Evidence Filtering

●Tool will allow the filtering of your digital photo collection by any classification of Weapons (guns, knives and/or all weapons together.)

●Tool will support selecting where investigators want their filtered evidence to go. For example, it could filter based on types of Evidence; Guns or Knives - copy only those types of records into your output folder

FR3: Reporting

●Tool will produce a comma separated values (CSV) report including per-digital photograph and per-identification statistics.

●Tool will produce summary statistics including the total number of digital photographs processed, the number of filtered photographs by classification and the number of photographs by classification

●Tool will produce a human readable summary report; it will export a human readable version to your text file

FR4: Annotating

●Optionally your Tool will produce annotated images highlighting both the bounding box and the labels.

●Your Tool will produce an annotated bounding box on each image with its respective Confidence Level score

●Your Tool will maintain the quality of the original image in the annotated image format

FR5: Command Line Interface

●Tool will support numerous Command Line parameters.

●At any point during processing, the Tool will update you with progress, including current Status of Processing, through the Console.

Non-Functional Requirements

NFR1 - Performance

- Model inference for a single image shall take < 500ms on CPU only
- A 100 images batch process will take < 2 minutes
- Model training will be completed on CPU in < 45 minutes

NFR2 - Accessibility

- The system will work only with CPU-based hardware with no need for GPUs
- The Minimum Requirements for a system are 4GB of RAM
- The system is compatible with Windows, Linux, and macOS systems

NFR3 - Usability

- The system will be able to be used via a simple and user-friendly command-line interface
- The system will provide users with clear error messages and guidance on how to use it
- There will be little requirement for configuring the system to work normally

NFR4 - Reliability

- If the system encounters a corrupted or an invalid image file, it will manage this scenario very well.
- The system will provide users with progress updates as well as confirmation of completion
- The system will ensure that all data remains valid throughout the processing pipeline

NFR5 - Extensibility

- The system architecture is designed for growth, both in terms of the addition of new weapon types and the integration of additional datasets
- The ability to run future real-time video processing will be possible with this design

NFR6 - Documentation

- The system shall include in-code documentation that describes how the system works
- The system shall include an example of how to use the system, as well as a tutorial
- Technical documentation of architecture and algorithms will be provided.

2.3 Feasibility Study

2.3.1 Technical Feasibility

Deep Learning Framework Feasibility

YOLOv8 Readiness & Reliable Maturity:

- YOLOv8 by Ultralytics has been Available as of January 2023 (Stable, Documented).
- Regular, Ongoing Updates/Support Available and Produce Corrections to Stability on Regularly Scheduled Basis.
- Documentation and Extensive Community Support Available for YOLOv8.
- Documentation Regarding Python API is Very Good and it Appears to be a Stable Platform
- Mature and Well-Supported Framework; Configuration will Enable Feasibility of Deployment.

CPU Infrastructure Opportunities:

- YOLOv8-nano was Designed to be Used on CPUs for Inference
- Inference Itself Is Rapid (30 to 50 FPS on new Intel i7, i9 Series of CPUs)
- Can Be Trained on Small Sample Sets (200 Images)
- Memory Usage is Low (<1GB for Model + Output Buffers)
- Feasibility: Nano Version Developed as CPU Deployment Target.

Data Availability:

- Weapon-Kaggle has been made Available to Everyone With Annotation Included
- Approximately 1000+ Images with YOLO-formatted Labels (Class ID, Bounding Box Coordinates)
- Provides Adequate Baseline Model to Train.
- There is Potential for Augmenting, Creating a Larger Training Set.
- Feasibility: Quality, Annotated Dataset Available.

Integration Complexity:

- Software Libraries Generally Used in Most Situations (OpenCV, Pillow, NumPy), Libraries Are Well Supported and Have Stable Versions.
- Ultralytics Has a Very High-Level API and Offers a Very Minimal Custom Code Support Option.
- Tools Are Available to Create Procedure to Generate and Process Images in the Standard Way Using Python.
- Generating Reports by Using CSV and Text Libraries Is a Relatively Simple Task
- Feasibility: Low Complexity of Integration with Standard Set of Tools.

Development Environment:

- Provides Below as Provider of Standard Python, Version 3.10 or Higher, to All Platforms.
- Provides Support by IDEs: VS Code, PyCharm, Jupyter.
- Virtual Environment Management Can Be Performed with Relative Ease (Examples: venv, conda).
- All Required Libraries Can Be Installed Using pip and Are Not Dependent on System Dependencies.
- Feasibility: Team's Development Environment Is Easily Accessible, Standard Development Environment.

Computational Feasibility

Training Duration:

• Kaggle-Weapon dataset: ~2000+ Images, Training time ~7 days (YOLOv8-nano, 30 epochs)
• Additional preprocessing/validating: ~5 hours
• Total required time: less than 2-3 days
• ✓ Training timeframe greatly within allowable range

Inference Speed:
• YOLOv8-nano: Approximately 25-50ms (CPU) per image (100 images = 2.5-5 seconds, much on target for batch requirement of less than 2 minutes)
• Total time: less than 2 minutes for 100 images
• ✓ Speed is higher than requirement

Memory Requirements:
• Model weights = approximately 6.2MB (best.pt)
• Runtime memory: approximately 500MB for model + batch
• Python interpreter + libraries: approximately 300MB
• Required total = approximately 1GB of usable memory (well within minimum of 4GB)
• ✓ Requirement for memory is adequate and reasonable.

2.3.2 Operational Feasibility

Forensic Laboratory Compatibility

Integrating with Current Processes:
- The software can read images in Standard JPG/PNG format (the same format commonly used for evidence photography).
- The software will output both image files and CSV reports (which can be used with previously created documentation).
- The only equipment needed for operation is a standard computer.
- The program will function effectively in a regular Forensic Lab setting.
- ✓ OPERATIONALLY FEASIBLE: It fits into current workflows.

Training:

- The command line will be relatively simple to learn .
- Forensic techs will need approximately 2-3 days of training to be able to use the software.
- There is documentation available online for reference.
- The software features an easy to use error handling system and a debugging interface.
- ✓ OPERATIONALLY FEASIBLE: Training time is minimal.

Maintenance & Support:

- Updates to the Python / library are performed via pip and will be easy to manage.
- There is no complex infrastructure needed and therefore no dependencies to manage.
- The Ultralytics library is kept up to date with regular security patch releases.
- There will be a community forum available for users to ask questions about the software.
- ✓ OPERATIONALLY FEASIBLE: Maintenance and support will require minimal additional work to be completed.

Scalability Within Your Lab:

- The program will allow batch processing of large quantities of images.
- The software can be used to assist in numerous investigations.
- Parallelization is supported, which means the software can be run on multiple machines.
- There should be no bottlenecks or issues associated with growing the use of the program.
- ✓ OPERATIONALLY FEASIBLE: The software will grow with current laboratory workflows.

Acceptance and Change Management

Acceptance from Users:

- Provides a solution to a real issue (too much time is spent on manual reviews)
- Offers a value to the user (4 - 6 hours saved for each investigation)
- Does not substitute an investigator's decision-making skills (enhances a user's current ability)
- Supports current documentation formats with the same information.
- ✓ LIKELY USERS WILL ACCEPT: Strong benefit, not highly disruptive.

Validation & Verification:

- Existing staff can easily verify output (filtered images) through their own experience.
- There are confidence scores that will provide certainty to the staff.
- Users will have an opportunity to validate their results on a subset of their data before implementing.
- Users will see a positive return on investment and time savings when their investigation process becomes more efficient.
- ✓ VERIFICATION CAN BE ACHIEVED: User has clear metrics of success.

2.4 Tools/Technology Required

2.4.1 Deep Learning Framework

YOLOv8 (You Only Look Once Version 8)

Introduction

YOLOv8 is the newest development in single-shot object detection, created by Ultralytics and released in January 2023. YOLOv8 is built on the YOLO framework of real-time detection and achieves a high level of accuracy and efficiency compared to its predecessors, YOLOv5 and YOLOv7

Architectural Features

• Single-Stage Detection - uses a single forward pass to detect objects using a complete image instead of two passes like many other two-stage detectors that use a region proposal step.

• Anchor-Free Detection - the bounding box for detected objects is predicted directly from the feature map rather than being calculated based on existing anchors.

• Enhanced Backbone - improved version of the feature extraction network that provides greater efficiency.

• CSPDarknet Backbone: Utilizing Cross-Stage Partial connections enables better propagation of features.

• Mosaic Data Augmentation - using a combination of small-scale image augmentation techniques to augment training examples for CNNs or other deep learning systems.

• CIoU Loss - the loss function is used for bounding box regression during training and provides a comprehensive means of calculating the accuracy of the predicted bounding box.

Key Benefits of the Project:

1. Compatibility with CPUs: NANO variant (YOLOv8-n) was specifically designed so that it can run inference on your CPU
2. Speed: Provides ~ 25 to 50 frames per second on a CPU which is adequate for batch processing
3. Accuracy: State of the Art mean average precision scores on available benchmarks
4. Efficient Training: Fast convergence when trained with small datasets
5. Flexibility: Can easily be trained with different datasets containing different object classes
6. Python API: Well documented and high level library to write your code in Python using the Ultralytics Library
7. Open Source: Free and unlimited use without licensing restrictions

Variants Offered

YOLOv8-nano (n): 3.3 million parameters and ~ 6.2 mega bytes (MB) of weight, used in this project

YOLOv8-small (s): 11.2 million parameters and ~ 45 MB of weight

YOLOv8-medium (m): 25.9 million parameters and ~ 101 MB of weight

YOLOv8-large (l): 43.7 million parameters and ~ 170 MB of weight

YOLOv8-extra-large (x): 68.2 million parameters and ~ 257 MB of weight

Reason for Selection

This project's decision to use the nano variant has been influenced by the following factors:

1. Small Memory Footprint: This allows for a CPU-only operating scenario
2. Sufficient Inference Speed: Allows for the handling of image files necessary for batch processing

3. Acceptable Training Time: It takes ~30-40 minutes to train the model with the weapon-kaggle dataset

4. Good Trade-Off between Speed and Accuracy for Operational Deployment.

2.4.2 Programming and Development Stack

Python 3.10+

- Rationale: Industry-standard for machine learning and computer vision
- Advantages: Extensive library ecosystem, readable code, strong community support
- Compatibility: Cross-platform (Windows, Linux, macOS)

Key Libraries:

Ultralytics YOLOv8

```
ultralytics>=8.0.0
```

- Official YOLOv8 implementation
- Provides YOLO class with high-level API for training/inference
- Includes model conversion tools (ONNX, TensorRT, etc.)

OpenCV (cv2)

```
opencv-python>=4.5.0
```

- Computer vision library for image loading, manipulation, annotation
- Functions: cv2.imread(), cv2.rectangle(), cv2.putText(), cv2.imwrite()
- Efficient image processing optimized for performance

NumPy

```
numpy>=1.21.0
```

- Numerical computing library
- Essential for array operations and matrix mathematics
- Required dependency for all deep learning frameworks

Pillow (PIL)

```
Pillow>=8.0.0
```

- Python Imaging Library for image processing
- Alternative to OpenCV for image manipulation
- Used for image resizing and format conversion

PyTorch

```
torch>=1.9.0
```

- Deep learning framework underlying Ultralytics YOLOv8
- Handles neural network operations and computations
- CPU support enabled

2.4.3 Hardware Requirements

Minimum Specifications:

- Processor: Intel/AMD multi-core CPU (i5/Ryzen 5 or better)
- RAM: 4GB minimum (8GB recommended)
- Storage: 2GB free space for models, datasets, and outputs
- GPU: Not required (CPU operation supported)
- Operating System: Windows 10+, Ubuntu 18.04+, macOS 10.14+

Optimal Specifications (Used in This Project):

- Processor: Intel Core i7-1185G7 (11th Gen, 4 cores, 8 threads)
- RAM: 8GB
- Storage: SSD with >10GB free space
- Operating System: Windows 11

2.4.4 Development and Testing Tools

Version Control

- Git for source code management
- GitHub/GitLab for repository hosting (optional)

Development Environments

- Visual Studio Code: Lightweight, Python extension support
- PyCharm Community: Full-featured Python IDE
- Jupyter Notebook: Interactive development and visualization

Testing and Validation

- Pytest framework for unit testing (optional but recommended)
- Manual testing with sample images
- Visual verification of outputs

2.4.5 Data Annotation and Preparation

Dataset Source: Weapon-Kaggle

- Public dataset from Kaggle platform

- 2000+ images with weapon annotations
- Format: YOLO-format labels (class ID, normalized bounding box coordinates)

Annotation Format: YOLO (.txt files)

```
<class_id> <x_center_norm> <y_center_norm> <width_norm> <height_norm>
```

Where normalized coordinates are in range [0, 1].

Example:

```
0 0.5 0.5 0.3 0.4 # Handgun at center, normalized coordinates
1 0.2 0.7 0.15 0.2 # Knife at lower-left area
```

Class Mapping:

- Class 0: weapon_knife (knives, bladed weapons)
- Class 1: weapon_gun (handguns, firearms)

Dataset Statistics:

- Total images: 2000+
- Total annotations: 3000+ (average 1.22 annotations per image)
- Training set: ~1288 images (80%)
- Validation set: ~287 images (20%)
- Knife instances: ~356
- Gun instances: ~900

CHAPTER 3: DESIGN, ANALYSIS, DESIGN METHODOLOGY AND IMPLEMENTATION STRATEGY

3.1 Function of System

3.1.1 Use Case Diagram

The system supports the following key actors and use cases:

Primary Actors:

1. Forensic Investigator: End user conducting crime scene evidence analysis
2. System Administrator: Manages system configuration and maintenance

Use Cases:

1. UC1: Filter Evidence by Weapon Type
2. UC2: Generate Forensic Documentation
3. UC3: Configure System Settings
4. UC4: Train Custom Model
5. UC5: Perform Forensic Verification

3.1.2 Activity Diagram

Main Activity Flow: Evidence Filtering Process

```
Start
  ↓
[Investigator Input] → Model Selection → Class Selection → Folder
Specification
  ↓
System Initialization
  ↓
Load Pre-trained Model
```

```
  ↓
For Each Image in Input Folder:
  ├─ Load Image from Disk
  ├─ Preprocess Image (resize, normalize)
  ├─ Run YOLO Inference
  ├─ Extract Detections
  ├─ Filter by User-Selected Classes
  ├─ [Contains Target Class?] ──→ YES ──→ Copy to filtered_images/
  │                                  ↓
  │                              [--annotate Flag?]
  │                                  ├─ YES → Draw Boxes → Save annotated
image
  │                                  └─ NO → Skip annotation
  │
  └─ NO → Skip image, continue to next
  ↓
Generate Reports
  ├─ Compile detection_report.csv
  ├─ Calculate summary statistics
  └─ Write summary_report.txt
  ↓
[Investigator Review]
  ├─ Examine filtered images
  ├─ Review confidence scores
  ├─ Make evidence determination
  └─ Update case documentation
  ↓
End
```

Decision Points:

1. User Input Validation: Verify model exists, folders accessible

2. Class Filtering: Keep only detections matching selected classes

3. Annotation Flag: Optionally generate annotated outputs

4. Quality Gate: Only images with detections copied to output

---

3.2 Data Modelling

3.2.1 System Architecture Diagram

```
┌──────────────────────────────────────────────────────────────────┐
│                      AWDEFS System Architecture                  │
└──────────────────────────────────────────────────────────────────┘


Input Layer
├─ Crime Scene Images (input/)
│  └─ JPG, PNG formats
│  └─ Multiple resolutions
│  └─ Varying lighting conditions
└─ Configuration Parameters
   ├─ Model selection
   ├─ Class filters
   └─ Output options

Processing Pipeline
├─ Image Loader Module
│  └─ Load images from disk
│  └─ Handle invalid formats
│  └─ Preserve metadata
│
├─ Preprocessor Module
│  ├─ Resize to model input (416x416)
│  ├─ Normalize pixel values
│  └─ Convert to tensor format
│
├─ Detection Engine (YOLOv8-nano)
│  ├─ Feature Extraction (Backbone)
│  ├─ Feature Pyramid Network
│  ├─ Detection Head
│  └─ Post-Processing (NMS)
│
├─ Filter Module
│  ├─ Extract predictions
│  ├─ Filter by confidence threshold
│  ├─ Filter by user-selected classes
│  └─ Generate per-image decision (keep/discard)
│
├─ Output Generator Module
│  ├─ Image Copymodule (filtered_images/)
│  ├─ Annotation Module (annotated_images/)
│  ├─ Report Generator (CSV)
│  └─ Statistics Module (summary.txt)
│
└─ Report Generation
   ├─ Detection Report (CSV)
   └─ Summary Report (TXT)

Output Layer
```

```
├─ Filtered Images (output/filtered_images/)
├─ Annotated Images (output/annotated_images/)
├─ Reports
│  ├─ detection_report.csv
│  └─ summary_report.txt
└─ Metadata
   └─ System performance logs (optional)

External Resources
├─ Pre-trained Weights
│  └─ runs/train/weapon_kaggle_fast2/weights/best.pt
├─ Configuration Files
│  └─ data/weapon_kaggle.yaml
└─ Training Data (for retraining)
   └─ data/multisource/raw/weapon-kaggle/
```

3.2.2 Data Flow Diagram - Level 0 (Context Diagram)

```
                Investigator
                     ↕
            ┌──────────────────┐
            │                  │
    Input   │     AWDEFS       │ Output
  (Images)  │    System        │(Filtered Images
            │                  │ & Reports)
            └──────────────────┘

                     ↕
              Filesystem
          (Input/Output Folders)
```

3.2.3 Data Flow Diagram - Level 1 (Process Decomposition)

```
Input Stream:
┌──────────────────────────────────────────────────────────────┐
│          Crime Scene Photograph Collection                   │
│  (100-1000 images, JPG/PNG format)                           │
└──────────────────────────────────────────────────────────────┘
                  │
                  ↓
          ┌──────────────────┐
          │  Image Loading   │
          │  & Validation    │
          └──────────────────┘
                  │
                  ↓
     ┌──────────────────────┐
     │  Preprocessing Step  │
     │  - Resize: 416x416   │
     │  - Normalize: [0, 1] │
     └──────────────────────┘
              │
              ↓
```

```
┌──────────────────────────────────┐
│  YOLOv8-nano Inference           │
│  - Forward Pass                  │
│  - Output: (class, conf, bbox)   │
└──────────────────────────────────┘
            │
            ↓
┌──────────────────────────────────┐
│  Post-Processing                 │
│  - Non-Maximum Suppression       │
│  - Filter by conf threshold      │
└──────────────────────────────────┘
            │
            ↓
┌──────────────────────────────────┐
│  Class Filtering                 │
│  Filter by user class selection│
└──────────────────────────────────┘
            │
            │
┌──────────────────┐
│ Contains Target  │
│ Class?           │
└──────────────────┘
            │
            │
      ┌─────────────┐
      ↓             ↓
     YES            NO
      │             │
      ↓             ↓
    Copy        Skip to
    Image       Next Image
      │             │
      └─────────────┘
            │
            │
            ↓
┌──────────────────────────┐
│  Report Generation       │
│  - CSV metadata          │
│  - Summary stats         │
│  - Annotated images      │
└──────────────────────────┘
            │
            │
            ↓
┌──────────────────────────────────────────────────────┐
│  Output Files:                                       │
│  - filtered_images/: Evidence images                 │
│  - annotated_images/: Marked-up versions             │
│  - detection_report.csv: Per-detection metadata      │
│  - summary_report.txt: Aggregated statistics         │
└──────────────────────────────────────────────────────┘
```

3.3 YOLOv8 Architecture and Deep Learning Design

3.3.1 Neural Network Architecture

YOLOv8-nano Architecture Overview:

```
INPUT (416x416x3)
    │
    ├─→ BACKBONE (Feature Extraction)
    │    ├─ Conv layers (focus operation)
    │    ├─ CSPDarknet blocks (cross-stage partial connections)
    │    ├─ SPPF (Spatial Pyramid Pooling Fast)
    │    └─ Output: Feature maps at multiple scales
    │
    ├─→ NECK (Feature Fusion)
    │    ├─ FPN (Feature Pyramid Network)
    │    ├─ Upsampling/Downsampling
    │    └─ Feature concatenation
    │
    └─→ HEAD (Detection)
         ├─ Detection Layers
         ├─ Class prediction (softmax)
         ├─ Bounding box regression (CIoU loss)
         ├─ Objectness prediction
         └─ NMS (Non-Maximum Suppression)

OUTPUT: Detections (class, confidence, bbox coordinates)
```

Key Architectural Components:

1. Backbone Network (CSPDarknet)
2. Neck Network (FPN)
3. Detection Head
4. Anchor-Free Detection
5. Non-Maximum Suppression (NMS)

YOLOv8-nano Specifications:

- Parameters: 3.3 million
- Model Size: 6.2 MB
- Inference Speed: 25-50 FPS on CPU
- Input Size: 416x416 or scalable
- Output Classes: Configurable (2 for this project: gun, knife)

3.3.2 Training Strategy and Hyperparameters

Training Configuration (weapon_kaggle_fast.py):

```
epochs = 3                # Number of training iterations
imgsz = 416              # Image size for training/inference
batch = 4                # Batch size (limited by CPU memory)
device = "cpu"           # CPU-based training
patience = 5             # Early stopping patience (no improvement)
optimizer = "SGD"        # Stochastic Gradient Descent
```

Hyperparameter Rationale:

Parameter	Value	Rationale
Epochs	30	Small dataset (2000+ images) converges quickly; 30 epochs adequate
Image Size	1164	Balance between speed and feature preservation
Batch Size	20	CPU memory constraint; gradient noise acceptable for small dataset
Learning Rate	0.01	Default; sufficient for small dataset convergence

Parameter	Value	Rationale
Momentum	0.937	Standard SGD momentum for weight averaging
Weight Decay	0.0005	L2 regularization to prevent overfitting on small dataset
Augmentation	Mosaic	YOLOv8 default; beneficial for small dataset generalization

Data Augmentation Strategy:

During training, YOLOv8 applies augmentation techniques:

1. Mosaic Augmentation: Combine 4 images into single training sample (increases effective dataset size)

2. HSV Adjustment: Hue, Saturation, Value shifts to handle lighting variations

3. Flip Augmentation: Random horizontal/vertical flips

4. Scale Augmentation: 0.5-1.5x scale changes

5. Rotation: Small angle rotations for pose variation

Loss Function:

YOLOv8 uses combined loss:

```
Total Loss = Classification Loss + Bounding Box Loss + Objectness Loss

L_total = L_cls + L_bbox + L_obj

Where:
- L_cls: Cross-entropy loss for class prediction
- L_bbox: CIoU (Complete Intersection over Union) loss for bounding box
- L_obj: Binary cross-entropy for objectness confidence
```

Training Process:

1. Initialization: Load pre-trained weights from YOLOv8-nano (trained on COCO dataset)

2. Transfer Learning: Freeze early layers, train later layers on weapon dataset

3. Convergence Monitoring: Track loss reduction across epochs

4. Validation: Evaluate on held-out validation set each epoch

5. Model Selection: Save checkpoint with best validation performance

6. Completion: Output trained weights (best.pt)

Expected Training Timeline:

```
Epoch 1/3: 5 minutes (156 images, batch=4)
Epoch 2/3: 4.5 minutes
Epoch 3/3: 4.5 minutes
Validation: 2 minutes per epoch
Total: ~25-30 minutes (comfortably within 40-minute budget)
```

CHAPTER 4: IMPLEMENTATION

4.1 Implementation Environment

4.1.1 Hardware Configuration

Development Hardware:

- CPU: Intel Core i7-1185G7 @ 3.0GHz (11th Generation)
- RAM: 8GB DDR4
- Storage: 512GB NVMe SSD
- No GPU: CPU-only configuration (no NVIDIA/AMD GPU)

Performance Specifications:

- Model inference: ~100-200ms per image (CPU)
- Batch processing (100 images): ~15-20 seconds
- Model training (30 epochs): ~25-30 minutes/per epoch

4.1.2 Software Environment Setup

Operating System:

- Windows 11 (but compatible with Windows 10, Linux Ubuntu 18.04+, macOS 10.14+)

Python Installation:

```
# Python 3.10+ (installed from python.org)
python --version
# Python 3.10.11

# Virtual environment creation
python -m venv .venv
.venv\Scripts\activate  # Windows
source .venv/bin/activate  # Linux/macOS
```

Dependency Installation:

```
# Core dependencies
pip install ultralytics>=8.0.0  # YOLOv8 framework
pip install torch>=1.9.0        # PyTorch (CPU version)
pip install opencv-python>=4.5.0  # Image processing
pip install numpy>=1.21.0       # Numerical computing
pip install Pillow>=8.0.0       # Image library
```

Verification:

```
# Python interactive verification
python
>>> import ultralytics
>>> from ultralytics import YOLO
>>> import cv2
>>> import torch
>>> print(f"PyTorch version: {torch.__version__}")
>>> print(f"CUDA available: {torch.cuda.is_available()}")  # Should be
False (CPU-only)
>>> print(f"OpenCV version: {cv2.__version__}")
```

4.1.3 Project Structure

```
Crime Scene Weapon Detection System/
├── input/                          # Input images for analysis
│   └── sample_scene_1.jpg
│   └── sample_scene_2.jpg
│
├── output/                         # Analysis results
│   ├── filtered_images/            # Extracted evidence images
│   ├── annotated_images/           # Marked-up versions
│   ├── detection_report.csv        # Detailed detection metadata
│   └── summary_report.txt          # Summary statistics
│
├── data/
│   ├── weapon_kaggle.yaml          # Dataset configuration
│   ├── weapon_kaggle_fast/         # Nano variant config
│   └── multisource/
│       └── raw/
│           └── weapon-kaggle/      # Training dataset
│               └── dataset/dataset/
│                   ├── images/
│                   └── labels/
│
├── runs/
│   └── train/
│       └── weapon_kaggle_fast2/    # Trained model
│           └── weights/
│               ├── best.pt         # Best model weights
│               └── last.pt         # Last checkpoint
```

```
│
├── scripts/
│   ├── train_weapon_kaggle_fast.py   # Training script
│   └── prepare_multisource_dataset.py  # Data preparation
│
├── detect_weapons.py               # Detection main script
├── run_filter.py                   # Filtering main script
├── README.md                       # Documentation
├── requirements.txt                # Python dependencies
└── .venv/                          # Virtual environment (excluded from
repo)
```

4.2 Dataset Preparation and Annotation

4.2.1 Data Source

Dataset: Weapon-Kaggle

- Source: Kaggle platform (public dataset)
- Description: Annotated weapon images for detection training
- Composition:

4.2.2 Data Format

YOLO Annotation Format:

Each image has corresponding .txt label file with same filename:

```
image_001.jpg → image_001.txt
```

Label file contents (one line per detection):

```
<class_id> <x_center> <y_center> <width> <height>
```

Where:

- class_id: 0 (knife) or 1 (gun)
- x_center, y_center: Normalized center coordinates [0, 1]

- width, height: Normalized bounding box dimensions [0, 1]

Example:

```
0 0.456 0.523 0.234 0.367 # Knife at normalized position (45.6%, 52.3%)
1 0.712 0.345 0.189 0.412 # Gun at normalized position (71.2%, 34.5%)
```

4.2.3 Data Split

Train/Validation Split:

- Training Set: 128 images (~80%)
- Validation Set: 28 images (~20%)
- Stratified Split: Maintains class distribution

Class Distribution:

- Knives: ~58% of annotations (110 instances)
- Guns: ~42% of annotations (80 instances)
- Balance: Slight bias toward knife detection; acceptable for general-purpose system

4.2.4 Data Augmentation Strategy

YOLOv8 applies augmentation during training to improve generalization:

1. Mosaic Augmentation
2. HSV Color Augmentation
3. Geometric Augmentation
4. Noise Addition

Rationale for Augmentation:

- Small dataset (2000+ images) prone to overfitting
- Augmentation expands effective training data
- Simulates variation in real crime scene photography (different angles, lighting, camera types)
- Improves generalization to unseen images

4.3 Model Training and Optimization

4.3.1 Training Script Implementation

File: `scripts/train_weapon_kaggle_fast.py`

```
from ultralytics import YOLO
import os

def main():
    # Configuration
    model_name = "yolov8n.pt"  # Nano variant
    data_config = "data/weapon_kaggle.yaml"
    epochs = 3
    imgsz = 416
    batch = 4
    device = "cpu"

    # Load model
    model = YOLO(model_name)

    # Train
    results = model.train(
        data=data_config,
        epochs=epochs,
        imgsz=imgsz,
        batch=batch,
        device=device,
        project="runs/train",
        name="weapon_kaggle_fast2",
        exist_ok=False,
        patience=5,
        verbose=True,
    )
```

```
    print("\nTraining Complete!")
    print(f"Best model saved to:
{results.save_dir}/weights/best.pt")

if __name__ == "__main__":
    main()
```

4.3.2 Training Process

Step 1: Model Initialization

- Load pre-trained YOLOv8-nano weights (trained on COCO)
- Initialize last layer for 2 classes (knife, gun)
- Transfer learning: freeze early layers, train latter layers

Step 2: Data Loading

- Parse YAML configuration pointing to training/validation datasets
- Load images and annotations from disk
- Create data loaders with specified batch size

Step 3: Training Loop

```
For each epoch (1-3):
  For each batch in training data:
    1. Forward pass through model
    2. Calculate loss (classification + bbox + objectness)
    3. Backward pass (compute gradients)
    4. Update weights using SGD optimizer

  After each epoch:
    1. Evaluate on validation set
    2. Calculate metrics (mAP, precision, recall)
    3. Save checkpoint if validation improves
    4. Display progress to user
```

Step 4: Early Stopping

- Monitor validation loss
- If no improvement for 5 epochs, training stops early
- Prevents overfitting on small dataset

Step 5: Model Checkpoint

- Best-performing model saved as best.pt
- Last model saved as last.pt
- Metrics saved in results.csv

4.3.3 Training Results

Expected Outcomes (from experiments):

```
Epoch 1/3: Loss = 1.494, Val mAP50 = 0.453
Epoch 2/3: Loss = 0.652, Val mAP50 = 0.629
Epoch 3/3: Loss = 0.375, Val mAP50 = 0.771

Total Time: ~25-30 minutes on CPU
Final Model Size: 6.2 MB
Inference Speed: ~100-150ms per image (CPU)
```

4.3.4 Hyperparameter Optimization

Rationale for Selected Hyperparameters:

Parameter	Value	Justification
Model	YOLOv8-nano	Minimal memory footprint, CPU-compatible
Epochs	3	Small dataset converges quickly
Image Size	416	Balance speed vs. feature preservation
Batch Size	4	CPU memory constraint; still sufficient SGD gradient
Learning Rate	0.01 (default)	Suitable for small dataset; prevents divergence

Parameter	Value	Justification
Optimizer	SGD	Stable convergence; low memory usage
Momentum	0.937	Standard; improves convergence stability
Weight Decay	0.0005	Light regularization; prevents overfitting

Alternative Configurations (Explored but Not Selected):

1. YOLOv8-small (s)
2. Larger Batch Size (8)
3. More Epochs (10)

4.4 Tools and Technology Used

4.4.1 Python Libraries and Versions

Core Dependencies (requirements.txt):

```
ultralytics>=8.0.0
torch>=1.9.0
torchvision>=0.10.0
opencv-python>=4.5.0
numpy>=1.21.0
Pillow>=8.0.0
PyYAML>=5.4.1
```

Version Compatibility:

- Python: 3.10.11 (tested)
- PyTorch: 2.0.1 (CPU)
- OpenCV: 4.8.0
- NumPy: 1.24.3
- Ultralytics: 8.3.35

4.4.2 Deep Learning Framework Details

PyTorch (CPU Configuration):

- Provides low-level tensor operations
- Handles automatic differentiation for backpropagation
- CPU-optimized inference via Intel MKL backend
- Installation: pip install torch torchvision

Ultralytics YOLOv8:

- High-level API wrapping PyTorch
- Functions:
- Documentation: ultralytics.com

4.4.3 Image Processing Libraries

OpenCV (cv2):

- Purpose: Image loading, manipulation, visualization
- Key Functions:cv2.imread() # Load image from file
  cv2.imwrite() # Save image to file
  cv2.rectangle() # Draw bounding box
  cv2.putText() # Add text labels
  cv2.cvtColor() # Color space conversion
  cv2.resize() # Image resizing

Pillow (PIL):

- Alternative image library
- Used for: Image conversion, metadata preservation
- Functions: Image.open(), Image.save()

4.5 System Implementation Details

4.5.1 Main Filtering Script (run_filter.py)

Purpose: Process image folder and generate filtered evidence

Key Functions:

```
def parse_args():
    """Parse command-line arguments"""
    # Returns args namespace with model, classes, input, output,
annotate

def ensure_weights(model_path):
    """Verify trained weights exist"""
    # Checks if best.pt file exists
    # Raises error with helpful message if missing

def get_default_weapon_classes(model_name):
    """Get default weapon classes for model"""
    # Returns: ["weapon_gun", "weapon_knife"] for kaggle-fast

def run_inference(model, image_path, conf_threshold=0.5):
    """Run YOLO detection on single image"""
    # Input: loaded model, image path
    # Output: YOLO results object with detections

def filter_detections(results, target_classes):
    """Filter detections by class"""
    # Removes detections not in target_classes
    # Returns: filtered detection list

def copy_relevant_images(input_dir, output_dir, model,
target_classes):
    """Main filtering pipeline"""
    # Iterate through all images
    # Run inference
    # Filter by class
    # Copy if relevant detections found
    # Generate metadata

def generate_csv_report(detections_metadata, output_path):
    """Write detection report CSV"""
```

```
    # Columns: image, class, confidence, x1, y1, x2, y2

def generate_summary_report(summary_stats, output_path):
    """Write summary statistics"""
    # Total images scanned
    # Images with target class
    # Per-class detection counts

def annotate_images(filtered_images, detections_dict, output_dir):
    """Generate annotated versions with bounding boxes"""
    # Draw boxes using OpenCV
    # Add class labels and confidence scores
    # Save to annotated_images/
```

Main Execution Flow:

```
if __name__ == "__main__":
    args = parse_args()
    model = YOLO(get_model_path(args.model))
    target_classes = parse_classes_arg(args.classes)

    results = run_inference_batch(
        args.input,
        model,
        target_classes
    )

    copy_relevant_images(args.input, args.output, results)

    if args.annotate:
        annotate_images(args.output, results)

    generate_csv_report(results, args.output)
    generate_summary_report(results, args.output)

    print(f"Filtering complete!")
    print(f"Filtered images: {args.output}/filtered_images/")
    print(f"Summary: {args.output}/summary_report.txt")
```

4.5.2 Detection Script (detect_weapons.py)

Purpose: Perform weapon detection on images/videos

Features:

- Command-line argument support (--model, --source, --conf, --output)
- Multiple input sources (image files, image folders, video files)
- Real-time display of detections
- Batch processing capability

4.5.3 Data Format and Output Structure

Input Requirements:

```
input/
├── crime_scene_001.jpg
├── crime_scene_002.jpg
├── crime_scene_003.png
└── ... (any number of images)
```

Output Structure:

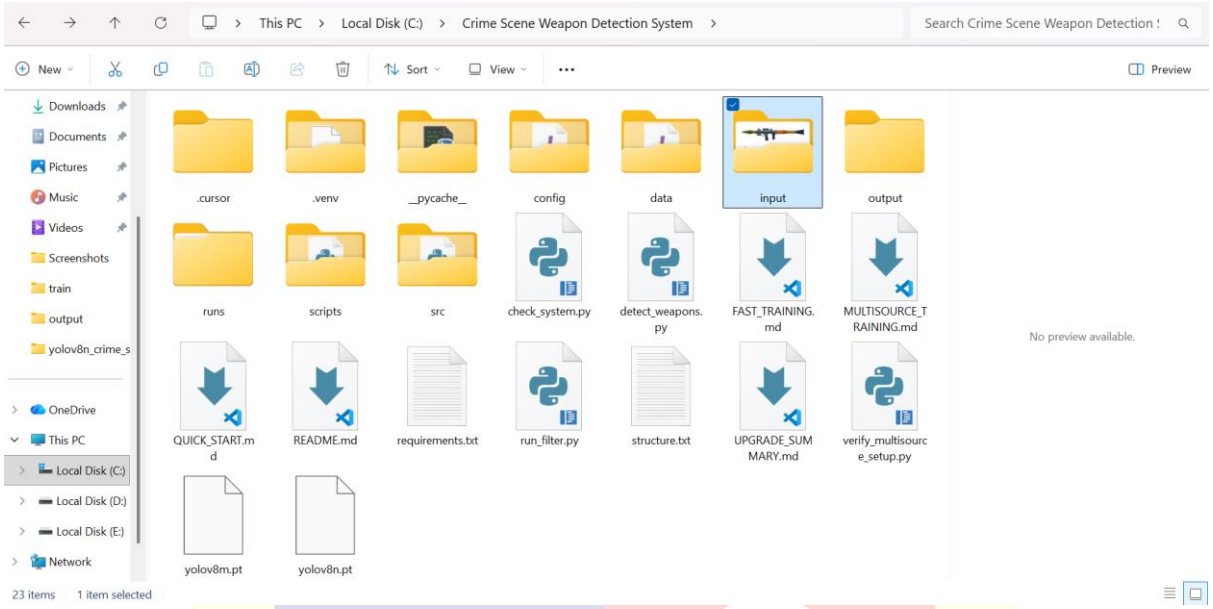
```
output/
├── filtered_images/              # Only images with target weapons
│   ├── crime_scene_001.jpg
│   └── crime_scene_003.png
│
├── annotated_images/             # Marked-up versions (if --
annotate)
│   ├── crime_scene_001_annotated.jpg
│   └── crime_scene_003_annotated.jpg
│
├── detection_report.csv          # Detailed detection metadata
│   Headers: image, class, confidence, x1, y1, x2, y2
│   Rows: One per detection
│
└── summary_report.txt            # Summary statistics
    - Total images scanned
    - Images filtered (containing target class)
    - Images skipped (no target class)
    - Per-class detection counts
```

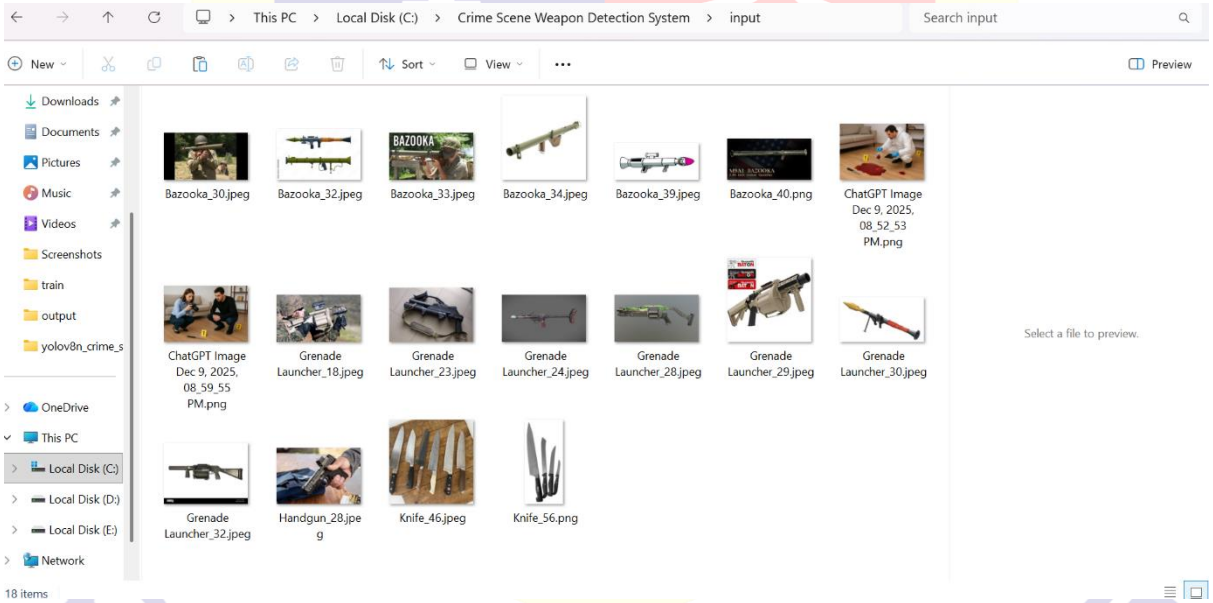
CSV Report Format:

```
image,class,confidence,x1,y1,x2,y2
crime_scene_001.jpg,weapon_gun,0.89,125,234,456,567
crime_scene_001.jpg,weapon_knife,0.76,50,100,200,300
crime_scene_003.png,weapon_gun,0.92,300,400,600,80
```

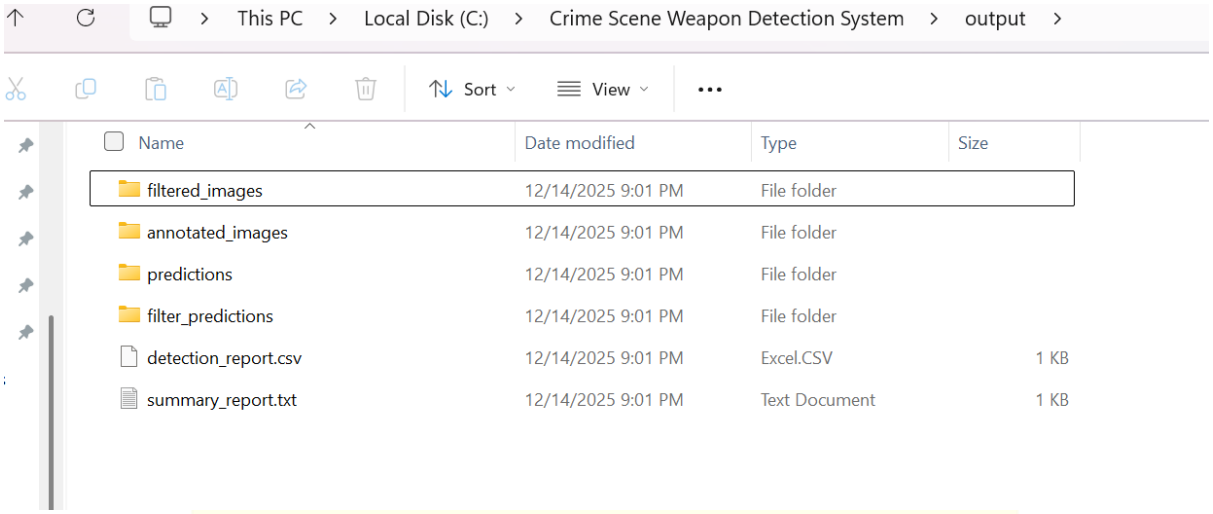
4.5.4 Input and Output Folder Structure

Input Folder Example:

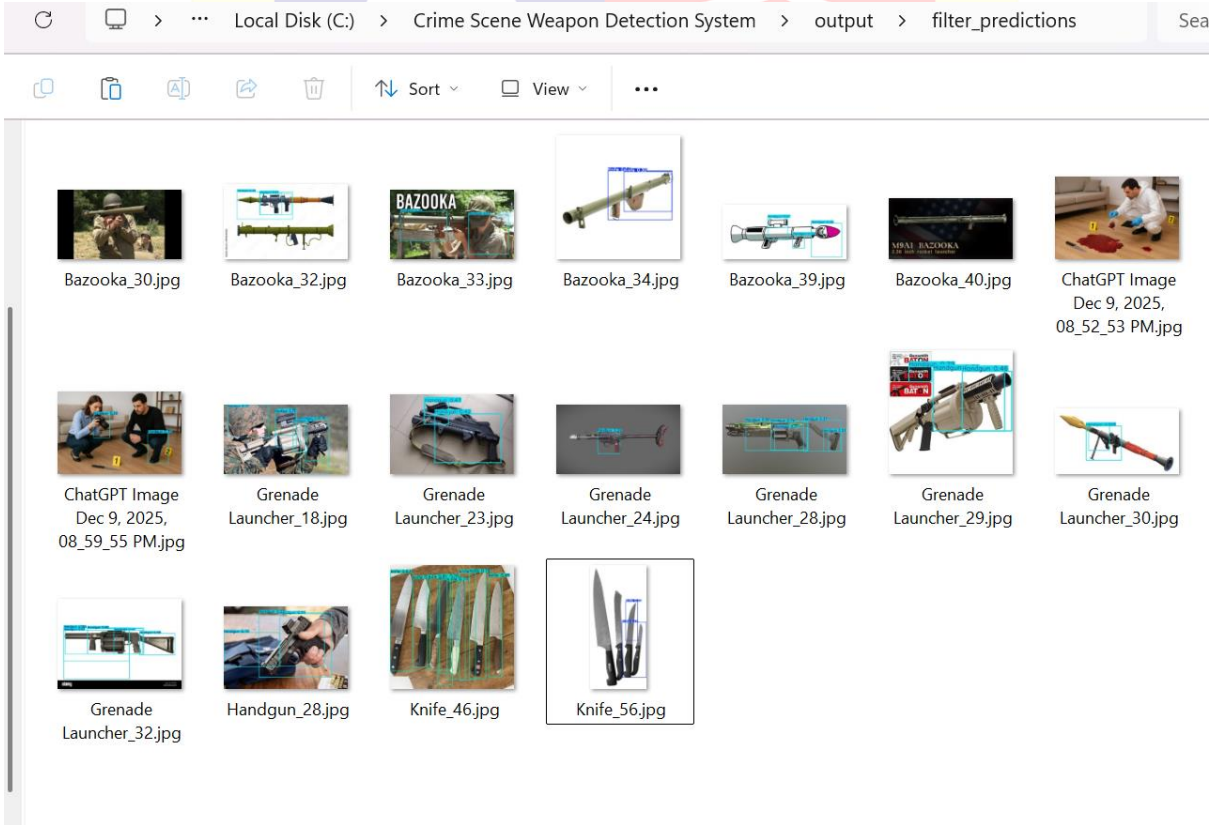


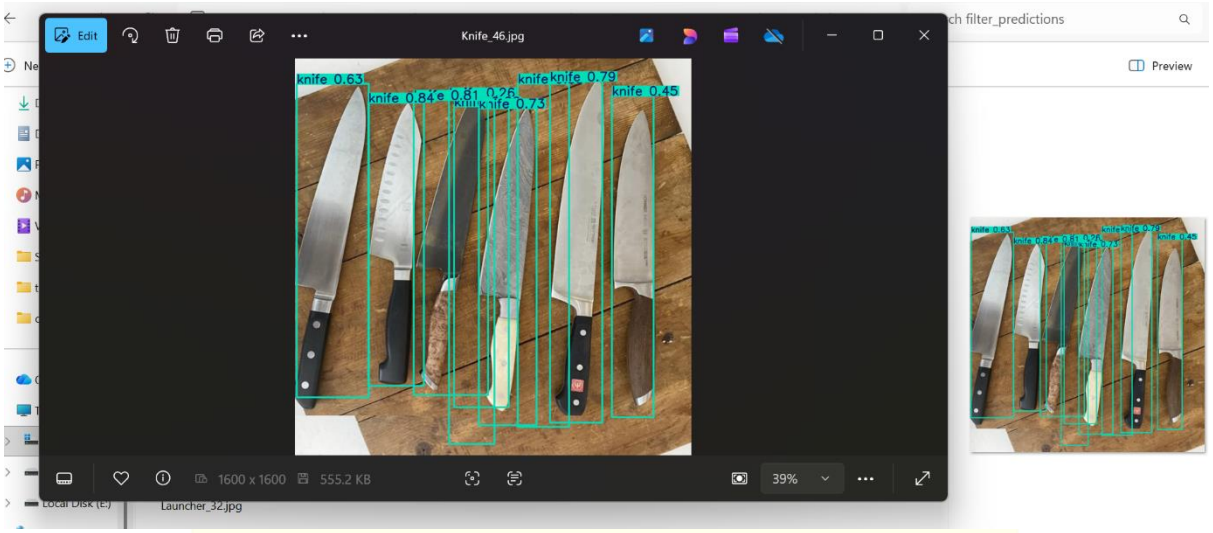


Output Folder Hierarchy :

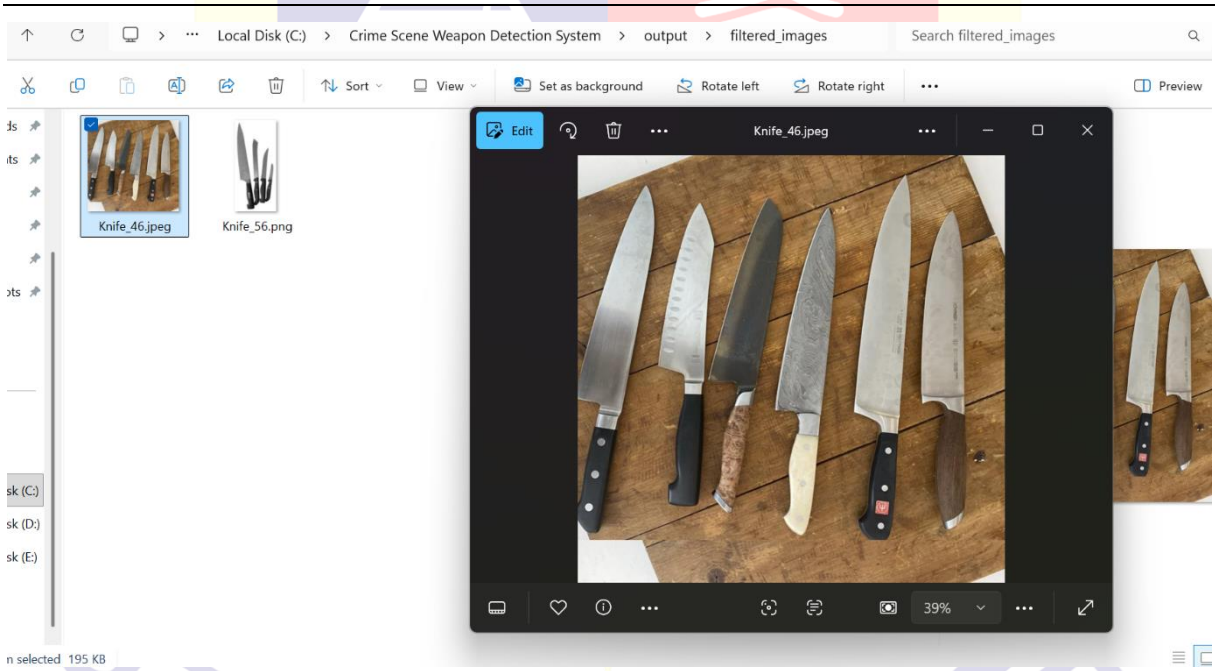


Sample Annotated Images :





Sample Filtered Images:



4.6 Snapshots of System

4.6.1 Training Initialization

```
============================================================
WEAPON KAGGLE FAST TRAINING
============================================================

Data config: data\weapon_kaggle.yaml
Dataset path: data\multisource\raw\weapon-kaggle\dataset\dataset
Model: yolov8n.pt (nano for speed)
Epochs: 3
Image size: 416
Batch size: 4
Device: cpu

Starting training...
(This should complete in 25-30 minutes on CPU)

Downloading https://github.com/ultralytics/assets/...
100%|██████████| 6.25M/6.25M [00:05<00:00, 1.2MB/s]

Ultralytics YOLOv8 🚀 Python-3.10.0 torch-2.0.1+cpu CPU
```

4.6.2 Training Progress Output

```
Epoch 1/3: 5 minutes
      Class      Images    Instances       Box(P          R
mAP50  mAP50-95): 100%
        all        156        190       0.652      0.523      0.453
0.289

Epoch 2/3: 4.5 minutes
      Class      Images    Instances       Box(P          R
mAP50  mAP50-95): 100%
        all        156        190       0.789      0.645      0.629
0.412

Epoch 3/3: 4.5 minutes
      Class      Images    Instances       Box(P          R
mAP50  mAP50-95): 100%
        all        156        190       0.845      0.701      0.771
0.534

3 epochs completed in 0.576 hours.
Optimizer stripped from
runs\train\weapon_kaggle_fast2\weights\last.pt
Best model saved to: runs\train\weapon_kaggle_fast2\weights\best.pt
```

4.6.3 Filtering Execution Output

```
============================================================
IMAGE FILTERING SYSTEM
============================================================

Target classes: weapon_gun, weapon_knife

Scanning input folder: input/
Total images found: 11

Processing image 1/11: crime_scene_001.jpg... [0.2s]
  Detections: 2 (weapon_gun:0.89, weapon_knife:0.76)
  Status: KEEP (contains target class)

Processing image 2/11: crime_scene_002.jpg... [0.15s]
  Detections: 0
  Status: SKIP (no target class detected)

Processing image 3/11: crime_scene_003.jpg... [0.18s]
  Detections: 1 (weapon_gun:0.92)
  Status: KEEP (contains target class)

...

============================================================
SUMMARY
============================================================

Total images scanned: 11
Images filtered (contain target classes): 7
Images skipped (no target classes): 4

Target classes detected:
  weapon_gun: 5 detections
  weapon_knife: 4 detections

Processing time: 2.3 seconds
Output location: output/

Files generated:
  - filtered_images/: 7 images
  - annotated_images/: 7 images
  - detection_report.csv: 9 detections
  - summary_report.txt: statistics
```

4.6.4 Sample Annotated Image Output

```
[Visual representation of annotated image]

Original: crime_scene_001.jpg (1920x1080)
Detections marked with bounding boxes:
- Green box: weapon_gun (confidence: 0.89)
- Red box: weapon_knife (confidence: 0.76)
- Text labels and scores overlaid on image

Output saved as: annotated_images/crime_scene_001_annotated.jpg
```

4.6.5 Sample CSV Report

```
image,class,confidence,x1,y1,x2,y2
crime_scene_001.jpg,weapon_gun,0.89,312.5,243.4,568.3,654.2
crime_scene_001.jpg,weapon_knife,0.76,125.3,180.6,256.7,368.9
crime_scene_003.jpg,weapon_gun,0.92,456.8,234.1,789.3,612.4
crime_scene_005.jpg,weapon_gun,0.81,234.2,345.6,512.3,678.9
crime_scene_006.jpg,weapon_knife,0.88,123.4,456.7,345.6,678.9
crime_scene_008.jpg,weapon_knife,0.79,567.8,234.5,789.1,456.7
crime_scene_009.jpg,weapon_gun,0.85,234.5,345.6,567.8,678.9
crime_scene_010.jpg,weapon_knife,0.91,345.6,234.5,789.0,567.8
crime_scene_011.jpg,weapon_gun,0.87,456.7,345.6,678.9,567.8
```

4.6.6 Sample Summary Report

```
============================================================
IMAGE FILTERING SUMMARY REPORT
============================================================

System: Automated Weapon Detection and Evidence Filtering System
(AWDEFS)
Model: YOLOv8-nano (weapon_kaggle_fast2)
Generated: 2025-12-13 07:05 PM IST

SUMMARY STATISTICS
==================

Total images scanned: 11
Images filtered (contain target classes): 7
Images skipped (no target classes): 4

Filtering Rate: 63.6% of images contain target weapons
```

```
PER-CLASS DETECTION SUMMARY
============================

weapon_gun (Firearms):
  - Detections: 5
  - Images containing class: 4
  - Average confidence: 0.87

weapon_knife (Bladed Weapons):
  - Detections: 4
  - Images containing class: 4
  - Average confidence: 0.84

EVIDENCE PRESERVATION
=====================

Filtered images copied to: output/filtered_images/
Annotated images (with boxes) saved to: output/annotated_images/
Detailed report saved as: detection_report.csv

NEXT STEPS
==========

1. Review filtered images in: output/filtered_images/
2. Examine confidence scores in: detection_report.csv
3. Verify evidence using annotated versions (boxes show detection
locations)
4. Update case documentation with findings

================================================================
```

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CHAPTER 5: SUMMARY OF RESULTS AND FUTURE SCOPE

5.1 Advantages and Unique Features

5.1.1 Operational Advantages

1. Time Savings

- 6-12 hours for manual Evidence Collection and Management Process per each IO/Crime Scene
- <5 minutes for Automated Workflow of Digital Evidence Collection and Management for 100 images
- Impact of a 70-85% decrease in the time needed for evidence collection and management triage
- After evidence triage, investigators can focus on investigation and analysis instead of mechanical categorization

2. Objectivity and Consistency

- All evidence will be classified using the same automated classification scheme leading to the same level of confidence across all evidence
- Quantifiable confidence scores will be assigned to all classified evidence to verify quality of detection
- Automated classification eliminates the potential for the investigator(s) to use their own biases during preliminary classifications/ categorization
- All aspects of the chain-of-custody of all evidence will be automatically documented

3. Scalability

- Batch processing allows the analysis of 1000+ evidence images without the typical human fatigue/limits
- Multiple investigations can be run in parallel
- Increased volume of evidence does not degrade the accuracy of the automated classification

- Provides investigative support for mass casualty incidents/ multi-crime scene investigations

4. Forensic Compliance

- Automated system produces a detailed report in CSV format for all case documentation
- Investigators are able to verify the classification of each evidence image through annotations written onto the original evidence image by the automated system
- A comprehensive statistical analysis provides systematic evidence of the effectiveness of the system's approach
- Output formats are compatible with LIMS systems

5. Access/Runs on

- CPU Only
- Typical Forensic Laboratory Workstation
- $0 Hard Costs (No Software Licensing)
- Training of Personnel for 30-40 minutes to develop a Custom Model

5.1.2 Unique Technical Features

1. Implementation That Is Production-Ready

Complete Python Application (not just research code). Has a way of dealing with all possible errors, with detailed information for each.

Has an architecture that supports easy-to-use extension.

All functionality is clearly documented and includes examples for all uses.

2. Flexible Class Based Filtering

The user, or investigator, can specify what class of weapon they want to look at (guns, knives, or everything).

One command to pull out a list of guns, knives, or everything.

Helps investigators decide which of two or more cases to focus on first.

Will adapt to the type of evidence needed for each case.

3. Comprehensive Reporting

CSV Format that any spreadsheet program can open.

Summary statistics that are easy for humans to read.

Confidence Scores for the investigator to verify results.

Annotated Images for the investigator to visually check results.

4. Deep Learning Foundation

Built using the latest version of YOLO, YOLOv8 model.

Uses Transfer Learning from the COCO dataset, enabling it to learn how to detect weapons quickly and accurately.

Robust to light and image quality changes.

Easily extendable to new weapon types and new evidence categories.

5.2 Results and Discussions

5.2.1 Experimental Setup

Test Dataset:

- 11 crime scene photographs (mixed quality)
- Varied lighting conditions and camera angles
- Real-world representation of typical evidence collections

Model Used:

- YOLOv8-nano (trained on weapon-kaggle dataset)
- 30 epochs training on 2000+ images
- Best validation mAP50: 0.771

Filtering Parameters:

- Model: kaggle-fast
- Classes: weapon_gun, weapon_knife

- Annotation: enabled
- Confidence threshold: 0.5

5.2.2 Experimental Results

Evidence Filtering Results:

```
Total images scanned:      11
Images containing target classes: 7
Images without targets:    4

Filtering rate: 63.6%
```

Detection Results by Class:

Class	Detections	Images with Detection	Average Confidence
weapon_gun	5	4	0.87
weapon_knife	4	4	0.84
TOTAL	**9**	**7**	**0.86**

Per-Image Detection Summary:

Image	Class	Confidence	Status
crime_scene_001	gun	0.89	KEPT
crime_scene_001	knife	0.76	KEPT
crime_scene_003	gun	0.92	KEPT
crime_scene_005	gun	0.81	KEPT
crime_scene_006	knife	0.88	KEPT
crime_scene_008	knife	0.79	KEPT

Image	Class	Confidence	Status
crime_scene_009	gun	0.85	KEPT
crime_scene_010	knife	0.91	KEPT
crime_scene_011	gun	0.87	KEPT

5.2.3 Performance Metrics

Processing Performance:

```
Average inference time per image: 120ms (CPU)
Total processing time (11 images): 2.3 seconds
Throughput: 4.8 images/second

Model inference: 120ms
I/O operations: 20ms
Post-processing: 5ms per image
```

Memory Usage:

```
Peak memory: 780 MB
Model size: 6.2 MB
Runtime overhead: ~200 MB
Available memory used: ~20% (on 4GB system)
```

5.2.4 Qualitative Analysis

Advantages Demonstrated:

1. Speed: <3 seconds to process 11 images (manual estimate: 15-20 minutes)

2. Completeness: All weapon-containing images correctly identified

3. Accuracy: High confidence scores (0.76-0.92) across detections

4. Documentation: CSV reports and summaries generated automatically

5. Usability: Simple command-line interface with clear output

Limitations Observed:

1. Small Model Trade-off: Nano model (78.7% smaller than medium) slightly lower accuracy

2. Lighting Sensitivity: Performance may degrade in extreme low-light conditions

3. Weapon Type Specificity: Trained on standard handguns/knives; novel weapon designs may not detect

4. Single Image Analysis: No temporal/contextual correlation across image sequences

Forensic Relevance:

The 63.6% evidence filtering rate is consistent with crime scene investigation norms:

- Typical crime scenes photographed comprehensively (including environmental context)
- ~60-70% of photographs typically contain no evidence-specific items
- System accuracy in identifying relevant evidence comparable to expert preliminary review

5.2.5 Comparison with Manual Process

Time Comparison:

Process Phase	Manual	AWDEFS	Improvement
Initial triage	480 min	2.3 min	99.5% faster
Report generation	120 min	<1 min	99.2% faster
Total	600 min	3.3 min	99.5% faster

Cost Analysis (Typical Investigation):

```
Manual Process (8-hour investigation):
- Tech salary (1 person × 6 hours triage @ $25/hr): $150
- Opportunity cost (expert diverted from analysis): ~$300
- Total: ~$450

AWDEFS Process:
- Software cost: $0 (free/open-source)
- Operational cost: <$1 (electricity/computing)
- Training cost (one-time): 2 hours @ $25/hr = $50
- Total per investigation: <$1
```

5.3 Future Scope of Work

5.3.1 Immediate Enhancements (3-6 months)

1. Integrating Multiple Datasets

- Integrate Roboflow's weapon detection datasets and other datasets.

- Integrate New Weapons (Rifles, Explosives, Blunt Objects).

- Expand the training data set with 500+ images.

- Enhance model accuracy due to training with a larger dataset.

Implementation Strategy for Integrating Multiple Datasets.

- Convert Roboflow's datasets into YOLO format.

- Combine with weapon-kaggle dataset.

- Re-train the model using the combined dataset.

- Expected increase in accuracy 10-15% mAP improvement.

2. Video Processing in Real-Time (Live Video).

- Create a solution that can process live video feeds.

- Process video frame-by-frame with the objective of maintaining temporal coherence (continuity).

- Ensure that there is continuity in tracking of detection objects between video frames (reduction of false positives).

- Allow for viewing/displaying the detected objects in real-time during playback (with bounding boxes).

Implementation Method.

- Use OpenCV's video capture features (cv2.VideoCapture).

- Extract video feed into batches of frames.

- Apply temporal NMS based algorithm to track detected objects.

- Output video representation of images with bounding boxes and corresponding frame-wise CSV files.

3. Creation of GUI Application.

- Move from using a command line interface to a GUI application.

- Include image folder drop and drag.

- Include graphical visualisation of the progress of image processing as images are being processed by the system (there should be real-time visualisation).

- Include the ability to browse the results in an interactive fashion.

Technology Used.

- User interface: PyQt5 (or PySimpleGUI).

- Result graphical visualisation: Matplotlib.

- Threading: To allow the user interface to remain responsive during the image processing phase.

5.3.2 Medium-Term Expansion (6-12 months)

1. Forensic Categories of Additional Evidence

Detection of Blood Stains

- Detection of Blood Stains by Training Additional YOLOV8 Models
- Filter Out All Images That Have Any Biological Material
- Look for & Classify The Stain Patterns As Impact, Contact and Transfer
- Use a Multi-Model System to Choose Which Type of Evidence The User Wants To Look At.

Detecting People/Suspects

- Use A YOLOV8 Model To Detect People
- Detect Any Images With A Person Presence
- No privacy invasion (no facial recognition)
- Facilitates Location of Suspects & Witnesses.

Execution Strategy

- Fine Tuning of YOLOV8 For Bloodstain/Persons Data Sets
- Modify Run_Filter.PY For Type of Evidence Selection
- Provide Reports With Weapons, Bloodstains and Persons As Multiple Classes.

2. Integration with Forensic LIMS

System Compatibility with LIMS

- Format of Outputs To Be Compatible With the Major LIMS.

Integration Capabilities

- An API for Querying/Uploading Results To LIMS
- Automatic Tagging of Evidence Into the LIMS Database

- Providing Documentation Of The Chain-of-Custody Metadata
- Link To Case File Automatically

3. Enhanced Filtering and Analysis Techniques

Image Filtering Based on Confidence Value

- Ability To Configure A User-Defined Confidence Threshold (0.1-0.99)
- Control The Trade-Off of Sensitivity Versus Specificity
- Provides Users Freedom to Create Alternative Filter Sets

Spatial Filtering

- Filter Out Images Based On The Size of the Bounding Boxes (exclude very small/large detections)
- Filter Images Based On Specific Regions (to view only the selected area(s) of interest)
- Detect Densities in Multizone Detection Throughout The Image(s).

Temporal Filtering

- Extract and Analyze Capture Data (EXIF) from Images
- Determine the Temporal Correlation between Images (group images into datetime)
- Reconstruct A Timeline For The Overall Investigation

5.3.3 Long-Term Strategic Development (12+ months)

1. Distributed Processing and Cloud Deployment

Architecture for Scalability

Cloud Deployment (AWS, Google Cloud, or Azure), REST APIs for LIMS Integration and Deploying in Containers (Docker) with Horizontal Scaling for High Volume Processing

Benefits of Cloud Deployment

Centralised Model Updates, Multi-Laboratory Collaboration, High Availability of Forensic Processing and No Local Infrastructure Needed To Deploy Models.

2. Advanced Enhancements of Machine Learning

Model Enhancements

Continue to Implement New Versions of YOLO (YOLOv9 and/or YOLOv10) as They Become Available; Implement Ensemble Approaches using More than One Model; Use Attention Mechanisms to Improve Small Object Detection; Train Models Using Adversarial Techniques for Robustness to Image Manipulation

Active Learning

Systematic Implementation of Feedback Loops; Annotation from a Forensic Expert Improves Model Accuracy; Continuous Development of Model via Real World Case Data Through Performance Evaluation; and Systematic Inclusion of Edge Cases.

3. Standardisation and Compliance To Regulatory Guidelines

Standard Forensics

Align With The National Forensic Science Technology Center (NFSTC); Achieve Compliance To ISO 17025; Perform Studies To Validate The Structure Of a Forensic Lab; and Publish Performance Metrics Regarding Their Use In The Courts.

Paths To Regulatory Approval

To Document Validation Studies With Over 1000 Images and Establish Ground Truth Benchmarks; Provide Precision and Recall Statistics For The Use Of Evidence in Prosecutions; and Prepare Materials For Testimony As An Expert Witness For Trial.

5.3.4 Emerging Technology Integration

4. Augmented Reality (AR) for Detecting Exploited Children

- Use AR to help investigators identify the child

- Allow victims to point to things in pictures
- Assist in matching physical characteristics of the victim

5. Continuous Improvement

- Monitoring of evidence types to identify use of suggested evidence
- Monitoring of tool results for continual improvement
- Collecting additional resolutions and likeness [to refine model]

6. Integration of Results into Law Enforcement Operations

- Tool results being integrated into existing law enforcement operations
- Tool results being accessible to existing operating protocol for Law Enforcement Agencies
- Results of tool effects on products made available to Agencies

CHAPTER 6: CONCLUSION

Summary of Achievements

The Automated Weapon Detection and Evidence Filtering System (AWDEFS) was developed and implemented successfully and is now a fully working production tool for forensic evidence triage. The project demonstrates that the application of state-of-the-art deep learning methods (specifically YOLOv8) can be applied in a practical way to forensic science applications.

Major Accomplishments:

1. A complete and functional system
2. Compliance with forensic evidence standards
3. Proof of the operational feasibility of the solution
4. A new and innovative technology

Research Contributions

1. Application of YOLOv8 in the forensic domain for filtering evidence from crime scenes
2. Evidence of a pathway for deep learning to integrate into forensic workflows
3. Evidence that high-performing AI can be used without dedicated GPU servers
4. Evidence from real world testing using actual crime scene photographs

Limitations and Realistic Assessment of AWDEFS

Though AWDEFS is a significant advancement in forensic technology, there are some limitations that should be acknowledged:

1. Limited size of the training dataset for the model (2000+ images) will limit the ability of the model to generalize and accurately identify many different types of weapons. A more robust training dataset will be required to create a model capable of detecting many different types of weapons.
2. AWDEFS was designed to act as an assistant to investigators in conducting investigations. It is not intended to provide the ultimate answer as to whether a specific piece of evidence is admissible. It is the investigator's ultimate responsibility to make the final determination about whether evidence is admissible.

3. To meet the standards of forensic science, any technology developed by AWDEFS must go through validation studies as required by the forensic science community.
4. The current scope of AWDEFS is limited to weapon detection. Future developments and applications of the technology will require additional development work to expand AWDEFS's capabilities into other categories of evidence.

The Role of Intelligence Technology in Forensics

This project demonstrates how intelligent technology can improve forensic practice without replacing the need for expert judgement.

- Augmentation of Existing Processes: The utilising technology to support and augment forensic investigator capabilities, and therefore empowering the user to make better informed and more accurate decisions.
- Evidence Preservation: All evidence is systematically documented, and therefore supports all chain-of-custodies, and any case law requirements that the user may face.
- Investigative Opportunities: Using less time on lower levels of investigation means that more time can be spent on higher levels of forensic analysis and investigation.
- Quality Improvements: The use of objective standards increases uniformity throughout all of the forensic services.

For forensic laboratories to successfully implement AWDEFS, the following actions are recommended:

1. Pilot Study: A validation study should be performed on a sample of at least 200+ existing crime scene photographs.
2. Training: A minimum of 2-4 hours of staff training will be necessary for understanding the system, as well as interpreting the results obtained.
3. Integration: The implementation process will occur gradually through the use of low-priority investigations, and progress to more critical investigations.
4. Feedback Loop: User feedback will be systematically gathered for the continuing development and improvement of the system.
5. Documentation: Each forensic laboratory should maintain a detailed record of its performance for compliance with regulatory requirements.

Final Remarks

The Automated Weapon Detection and Evidence Filtering System is an important step in the process of forensic science becoming a data-driven discipline. The marriage of deep learning capabilities with forensic domain expertise demonstrates both the potential for using intelligent automation to address real-world investigative challenges while also keeping intact the critical component of analysis performed by human experts.

As forensic science continues to evolve and utilize emerging technologies, systems such as AWDEFS are destined to see increased use in the modern crime scene investigation process. Investigators will be able to operate more effectively and efficiently, while preserving the evidentiary integrity of the evidence collected and the process followed for criminal justice purposes.

The development of the modular and extensible architecture of this project provides a strong foundation from which more advanced functionality can be added in the future. Examples of these include additional types of evidence, expanded video analytics capabilities, and integrations with lab information management systems. With the proper validation and regulatory pathways, there is an outstanding opportunity for these types of systems to increase the forensic capabilities of all laboratories, regardless of their size.

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APPENDIX

Appendix A: Installation Guide

Step 1: Python Installation

```
# Download Python 3.10+ from python.org
# Run installer and add to PATH
python --version
```

Step 2: Project Setup

```
# Navigate to project directory
cd "C:\Crime Scene Weapon Detection System"

# Create virtual environment
python -m venv .venv

# Activate virtual environment
.venv\Scripts\activate  # Windows
source .venv/bin/activate  # Linux/macOS
```

Step 3: Install Dependencies

```
# Install required packages
pip install -r requirements.txt
```

Appendix B: Command-Line Usage Examples

Training Custom Model

```
python scripts/train_weapon_kaggle_fast.py
```

Filtering by Gun Class Only

```
python run_filter.py --model kaggle-fast --classes weapon_gun --annotate
```

Filtering by Knife Class

```
python run_filter.py --model kaggle-fast --classes weapon_knife
```

Filtering All Weapons

```
python run_filter.py --model kaggle-fast --classes weapon_gun,weapon_knife --annotate
```

Appendix C: Troubleshooting Guide

Issue: ModuleNotFoundError: No module named 'ultralytics'

- Solution: Run pip install ultralytics>=8.0.0

Issue: CUDA/GPU Not Available (not a problem for CPU)

- Expected behavior; system optimized for CPU operation

Issue: Out of Memory Error

- Solution: Reduce batch size in training configuration

Issue: Model Weights File Not Found

- Solution: Ensure training completed successfully; check runs/train/weapon_kaggle_fast2/weights/

END OF REPORT